

Swaption model 1 into 3						
Time	Swap rate	zc	Period	DF(0, t)		
1	5.000%	5.000%	0x1		d_1	0.2568
2	5.110%	5.113%	1x2	0.9051	d_2	0.0568
3	5.300%	5.312%	2x3	0.8562		
4	5.470%	5.494%	3x4	0.8074		
F	5.644%		$A =$	2.5687		
σ	20.0%				$XN(-d_2) - FN(-d_1)$	0.00361
T	1		Notional	1,000,000	Receiver swaption	9,271
X	5.470%				(in bps)	92.7

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	(54,700)	(52,095)	50,000	47,619
2	5.113%	5.226%	(54,700)	(49,508)	52,257	47,297
3	5.312%	5.712%	(54,700)	(46,833)	57,122	48,907
4	5.494%	6.041%	(54,700)	(44,165)	60,414	48,778
				(192,601)		192,601
	Receiver strike		5.470%		Receiver	9,271
	Fixed rate		5.470%			
	Notional		1,000,000		NPV	9,271

Figure 6.4: Computing the Value of the 1 into 3 5.47% Receiver and Incorporating That Value into the Cancelable Swap Model

embedded receiver swaption with the same strike. As evident with the few iterations we have done here, each successive time we have had to raise the fixed coupon by smaller and smaller amounts. Eventually we will come to a point where we don't need to move the coupon any higher, and this rate by definition is the cancelable swap rate.

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	(57,333)	(54,603)	50,000	47,619
2	5.113%	5.226%	(57,333)	(51,891)	52,257	47,297
3	5.312%	5.712%	(57,333)	(49,087)	57,122	48,907
4	5.494%	6.041%	(57,333)	(46,291)	60,414	48,778
				(201,872)		192,601
	Receiver strike		5.470%		Receiver	9,271
	Fixed rate		5.733%			
	Notional		1,000,000		NPV	0

Figure 6.5: Raising the Fixed Rate Until the Net Present Value of the Swap and the Embedded Receiver Swaption Equals Zero

Table 6.2 shows this procedure.⁸ The cancelable swap rate is 5.913%.⁹ Note that at a cancelable swap rate of 5.913% the negative value of paying fixed in a 4-year swap with a coupon of 5.913% is exactly offset with the value of a swaption with a strike of 5.913%. The fixed rate 5.913% in the cancelable swap is $5.913\% - 5.47\% = 44.3$ bps higher than the plain vanilla 4-year swap rate, representing a nice “pickup” over a regular swap. This pickup has of course been achieved by the fixed rate receiver by selling the embedded receiver swaption.

How does the pickup over the plain vanilla swap rate vary as we move the *lockout period*, or the period of time during which the swap cannot be canceled? Table 6.3 shows a variety of cancelable swaps all with the same final

⁸Thankfully, the advent of spreadsheet software makes the tedious process of manually iterating for the cancelable swap rate as in Table 6.2 unnecessary. One can link the cancelable swap rate with the strike of the embedded swaption and use a canned iteration function in the software package to solve for that rate that makes the trade an NPV = 0 transaction. Such a luxury has not always existed, however. There was a time in the industry when some people’s swap models and swaption models did not always “talk” to one another (e.g., one was built in a spreadsheet environment and the other was built in a C++ environment), and using a canned iteration function was not possible. In this case, one would have to iterate by hand to find the cancelable swap rate, much as we have done here. Disaster would strike when, upon completing the manual iteration process, it was necessary to change one of the input parameters to the model, requiring yet another bout of manual iteration.

⁹Once again, we assumed a flat skew and used an implied lognormal vol of 20% in Black’s formula to price up each of these swaptions. If we had considered the possibility of skew, we would have had to use a different implied vol in each step of the iteration, as the strike of the embedded option changed.

Swaption model		1	into	3		
Time	Swap rate	zc	Period	DF(0, t)		
1	5.000%	5.000%	0x1		d_1	0.0217
2	5.110%	5.113%	1x2	0.9051	d_2	-0.1783
3	5.300%	5.312%	2x3	0.8562		
4	5.470%	5.494%	3x4	0.8074		
F	5.644%		$A =$	2.5687		
σ	20.0%				$XN(-d_2) - FN(-d_1)$	0.00499
T	1		Notional	1,000,000	Receiver swaption	12,819
X	5.733%				(in bps)	128.2

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	(57,333)	(54,603)	50,000	47,619
2	5.113%	5.226%	(57,333)	(51,891)	52,257	47,297
3	5.312%	5.712%	(57,333)	(49,087)	57,122	48,907
4	5.494%	6.041%	(57,333)	(46,291)	60,414	48,778
				(201,872)		192,601
	Receiver strike		5.733%		Receiver	12,819
	Fixed rate		5.733%			
	Notional		1,000,000		NPV	3,548

Figure 6.6: Valuing the Receiver with a Strike of 5.733% and Incorporating This Value in the Cancelable Swap

maturity, allowing the lockout period to vary. As this table demonstrates, the shorter the lockout period, the greater the pickup is in the cancelable swap rate over the benchmark Treasury rate.

What's the intuition behind this result? The augmented fixed rate that is received by the fixed rate receiver compensates him for selling the option to cancel to the fixed rate payer. Instead of receiving an upfront premium for the sold option, the fixed rate receiver is being paid by receiving a higher

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	(58,341)	(55,563)	50,000	47,619
2	5.113%	5.226%	(58,341)	(52,803)	52,257	47,297
3	5.312%	5.712%	(58,341)	(49,950)	57,122	48,907
4	5.494%	6.041%	(58,341)	(47,104)	60,414	48,778
				(205,420)		192,601
	Receiver strike		5.733%		Receiver	12,819
	Fixed rate		5.834%			
	Notional		1,000,000		NPV	0

Figure 6.7: Again Raising the Fixed Rate Until the Net Present Value of the Swap and the Embedded Receiver Swaption Equals Zero

Iteration	Proposed Cancelable Swap Rate	Value Swaption	Value Swap	NPV
1	5.470%	9,271	0	9,271
2	5.733%	12,819	(9271)	3,548
3	5.834%	14,340	(12,819)	1,521
4	5.877%	15,019	(14,340)	679
5	5.897%	15,327	(15,019)	308
6	5.905%	15,468	(15,327)	141
7	5.909%	15,532	(15,468)	65
8	5.911%	15,562	(15,532)	30
9	5.912%	15,576	(15,562)	14
10	5.912%	15,582	(15,576)	6
11	5.913%	15,585	(15,582)	3
12	5.913%	15,586	(15,585)	1
13	5.913%	15,587	(15,586)	1
14	5.913%	15,587	(15,587)	0

Table 6.2: Iterating to the Cancelable Swap Rate

Structure	Embedded Swaption	Coupon
4nc1	1 into 3	$T_4 + 72.3$
4nc2	2 into 2	$T_4 + 56.1$
4nc3	3 into 1	$T_4 + 41.8$
4-year swap	n.a.	$T_4 + 28.0$

Table 6.3: Comparison of Cancelable Swap Rates in Our Stylized Example as a Function of Lockout Period. Swap Curve as in Table 6.1 and Implied Swaption Vol Assumed to Be 20% in Each Case

fixed coupon than is available in a plain vanilla swap. The only catch for the fixed rate receiver is that if the swap is canceled, then he stops receiving this high fixed coupon. With a very short lockout period, the fixed rate receiver may potentially have a short time to recoup his premium before the swap is canceled. In order to compensate him for this shorter period of time to essentially recoup his premium, this will mean that the pickup that he receives will have to be increased.¹⁰

6.2.3 Bermudan Cancelables

A *Bermudan cancelable swap* is a swap in which the fixed rate payer has the right to cancel the swap after some lockout period and periodically thereafter. Note that the exercise style is not American in nature: the fixed rate payer does not have the right to cancel continuously after some specified date but rather on multiple discrete dates. The fixed rate on a Bermudan cancelable can be no lower than on an analogous European cancelable.¹¹

¹⁰We should be a little careful with this way of thinking since the expiry, tail, strike, and ultimately the value of the embedded swaption changes as we move from one cancelable swap structure to another. But nonetheless this dynamic is at play as we move to shorter lockout periods.

¹¹By analogous we mean that the Bermudan and the European have the same final maturity and that the cancellation date of the European is the first cancellation date of the Bermudan. Suppose the fixed rate on a Bermudan cancelable swap were lower than the fixed rate on an analogous European cancelable swap. Then we could pay fixed on the Bermudan and receive fixed on the European thereby earning positive cash flow until the cancellation date of the European. If the European is cancelled then we should cancel the Bermudan as well, which means we will earn zero cash flow for the remaining life of the trade. If the European is not cancelled then we should not cancel the Bermudan either at this nor any subsequent cancellation date. This would lock in positive cash flow for the remainder of the trade. This then would be an arbitrage and thus the fixed rate on a Bermudan cancelable swap can be no lower than on an analogous European cancelable swap.

Callable Structure	Coupon Differential (in bps)
2nc6mo	6.25
2nc1	2.50
3nc6mo	13.75
3nc1	6.50
4nc6mo	22.00
4nc1	12.25
4nc2	4.00
5nc6mo	29.50
5nc1	17.50
5nc2	6.50
5nc3	2.25
7nc6mo	42.75
7nc1	27.25
7nc2	12.50
7nc3	5.50
10nc6mo	56.00
10nc1	38.50
10nc2	20.50
10nc3	12.00
10nc5	4.25

Table 6.4: Differential Between Fixed Coupons in Bermudan Cancelable Swaps and European Cancelable Swaps on a Trading Day in August 2005

Table 6.4 shows the differential (according to a propriety pricing model) in fixed rate coupons between Bermudan cancelables and European cancelables for a number of different structures. For example, on the date that this data was taken in August 2005, the fixed coupon for a 5nc1 Bermudan cancelable swap was 17.5 bps greater than that obtained in a 5nc1 European cancelable swap. These differentials will vary from day to day (or moment to moment), depending in part on the swap curve and vol grid prevailing, not to mention on the type of models used to price these trades.

Bermudan cancelables are popular structures, often used to swap out callable bond issuance as in Figure 6.2 where the bond is callable on more than one occasion. This type of call feature is common among many callable bonds, as it provides a higher coupon to the investor than an analogous bond that is callable on only one occasion.

The analysis of Bermudan cancelable swaps is complicated, much as is the analysis of Bermudan swaptions that we looked at in Chapter 5.¹² Even the decision to cancel a Bermudan cancelable on one of its multiple cancellation dates is nontrivial, much in the same way that the exercise decision of a Bermudan swaption can be complicated. It has not been unheard of to have sophisticated counterparties disagree on the decision to cancel a Bermudan cancelable swap on a particular date.¹³

Example 1

Consider again our stylized example and suppose that we have a 3nc1 Bermudan cancelable swap on the books in which we are paying fixed at a rate of 5.40%. This is a 3-year swap in which we have the right to cancel in one year or again one year after that. If we do not cancel the swap in a year, then we will be left paying fixed for two years with the right to cancel the following year (which means we will be left paying fixed in a 2nc1 single European in a year). That is to say, if we do not cancel the swap in a year, then we will be left paying fixed for two years at a rate of 5.40%, and we will be long a 1 into 1 receiver struck at 5.40%. The fact that we can cancel in a year means we are also implicitly long a 1 into 2 receiver struck at 5.40%.

Suppose now that one year has gone by. Suppose further that the swap curve is as given in Table 6.1 and that 1 into 1 swaption vol is 20%. Should we cancel the swap?

If we do not cancel the swap, the remaining portion of the trade will have us paying fixed at 5.40% for two years, with the right to cancel in one year. That is, if we do not cancel, we will be paying fixed in a 2nc1 single European cancelable swap. There are a couple of ways to approach this. The first way is to simply model up the residual trade: we should cancel if it has a negative NPV, and we should not cancel if it has a positive NPV. In the trade at hand, if we do not cancel, we will be paying fixed at 5.40% in a 2nc1. As shown in Figure 6.8, we would be better off canceling the trade since it has a negative NPV equal to $-\$71,655$, whereas if we cancel, we have no further cash flows.

Although looking at the NPV of the residual trade on a cancellation date is unequivocally the appropriate way to determine whether or not to exercise early, we can arrive at the correct exercise decision in another

¹²A Bermudan cancelable swap can be decomposed into a plain vanilla swap and a Bermudan receiver swaption, much in the same way a (single European) cancelable swap can be decomposed into a plain vanilla swap and a single European receiver swaption.

¹³See Longstaff, Santa-Clara, and Schwartz (2001b) for a discussion of how the decision to exercise a Bermudan swaption can depend very much on the type of model one employs.

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	(5,400,000)	(5,142,857)	5,000,000	4,761,905
2	5.113%	5.226%	(5,400,000)	(4,887,451)	5,225,748	4,729,739
3	5.312%	5.712%	0	0	0	0
4	5.494%	6.041%	0	0	0	0
5	5.635%	6.202%	0	0	0	0
(10,030,308)						9,491,644
Receiver						467,010
Fixed rate		5.400%				
Notional		100,000,000		NPV		(71,655)

Swaption model		1	into	1		
Time	Swap rate	zc	Period	DF(0, t)		
1	5.000%	5.000%	0x1		d_1	-0.0640
2	5.110%	5.113%	1x2	0.9051	d_2	-0.2640
3	5.300%	5.312%	2x3			
4	5.470%	5.494%	3x4			
5	5.600%	5.635%	4x5			
F	5.226%		$A =$	0.9051		
σ	20.0%				$XN(-d_2) - FN(-d_1)$	0.005160
T	1		Notional	100,000,000	Receiver swaption	467,010
X	5.400%				(in bps)	46.7

Figure 6.8: The Value of a 2nc1 with a Fixed Coupon of 5.40% when $\sigma = 20\%$

manner. In Figure 6.9, we solve for the par rate in a 2nc1 single European cancelable swap. This is a 2-year swap in which we pay fixed and have the right to cancel in one year. As shown in this figure, the par rate is 5.346%. If we do not cancel our swap today, we will continue to pay fixed at 5.40% in what amounts to be a 2nc1 single European. Since the fixed

SWAPS WITH EMBEDDED OPTIONS

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	(5,345,697)	(5,091,140)	5,000,000	4,761,905
2	5.113%	5.226%	(5,345,697)	(4,838,302)	5,225,748	4,729,739
3	5.312%	5.712%	0	0	0	0
4	5.494%	6.041%	0	0	0	0
5	5.635%	6.202%	0	0	0	0
						(9,929,442)
						9,491,644
						Receiver 437,798
Fixed rate			5.346%			
Notional			100,000,000	NPV		
						(0)

Swaption model		1	into	1		
Time	Swap rate	zc	Period	DF(0, t)		
1	5.000%	5.000%	0x1		d_1	-0.0135
2	5.110%	5.113%	1x2	0.9051	d_2	-0.2135
3	5.300%	5.312%	2x3			
4	5.470%	5.494%	3x4			
5	5.600%	5.635%	4x5			
F	5.226%		$A =$	0.9051		
σ	20.0%				$XN(-d_2) - FN(-d_1)$	0.004837
T	1		Notional	100,000,000	Receiver swaption	437,798
X	5.346%				(in bps)	43.8

Figure 6.9: The Par Rate in a 2nc1 when $\sigma = 20\%$

rate we are paying is above the par rate in a 2nc1, we should cancel the swap.¹⁴ □

¹⁴What if we want to maintain a position in the cancelable swap despite our rational exercise decision? If we want to maintain a position in a 2nc1, we would be better off canceling our trade with the 5.40% coupon and going into the market and entering into a 2nc1 at the par rate of 5.346%.

Example 2

Let's consider a slightly different scenario now. Suppose again a year has gone by and the swap curve is as in Table 6.1 just like before, but now suppose that the prevailing 1 into 1 swaption vol is now 25%, instead of 20%. Thus the only thing different between Example 1 and Example 2 is that the prevailing 1 into 1 swaption vol is now higher. Does our answer change?

Yes, it does. In this case, with the higher 1 into 1 swaption vol prevailing, we are better off not canceling in a year. We can again come to this realization in two ways. First, if we compute the NPV of the 2nc1 with the higher vol, we get a positive NPV equal to \$22,623 as shown in Figure 6.10. This indicates that we should not cancel the swap.

Alternatively, if we solve for the par rate in a 2nc1 with the higher vol assumption, we get a rate of 5.417%, as shown in Figure 6.11. Thus, since we can continue to pay at a rate of only 5.40% and the par rate is 5.417%, it is in our advantage not to cancel.¹⁵ $\square$

These two examples collectively show us that our decision to cancel or not in a year depends not only on the then-prevailing level of rates but also on the then-prevailing level of 1 into 1 swaption vol. With the lower level of vol, the remaining embedded option is not worth enough to offset the negative value of the plain vanilla swap, and we are better off canceling the trade. With the higher level of vol, this embedded option is very valuable, and we are better off not canceling the trade.

So what is the value of the 3nc1 Bermudan today?¹⁶ It's complicated. As we have just seen, the decision as to whether or not to cancel in a year depends in part on the then-prevailing level of 1 into 1 swaption vol. The higher that we think that 1 into 1 swaption vol will be in a year, the higher the value of the Bermudan 3nc1 today. We will return to this concept in Chapter 9.

¹⁵But what should we do if we just no longer want to maintain exposure in this trade, even though it is rational for us to not cancel the trade? In this case, if we want out of the trade, we should unwind the position instead of canceling the trade. The trade has a positive NPV to us, and we would get paid something to unwind it.

¹⁶Recall that we said that we have the 3nc1 Bermudan with the 5.40% coupon on our books today. We did not say that it was an NPV = 0 transaction. It could have been that we entered into this transaction a year ago when it was originally a 4nc1 Bermudan cancelable, and now a year has gone by and we have determined not to cancel the trade today. In fact, if we now assume that the swap curve today is also as given in Table 6.1 and that 1 in 2 swaption vol is now 20%, we can conclude that we would definitely not cancel our Bermudan trade today. Under these assumptions it can be shown that the par rate in a 3nc1 single European cancelable swap is 5.654%. Thus the par rate in a 3nc1 Bermudan cancelable must be at least 5.654%. Since we are currently paying a lower rate of 5.40% in our 3nc1 Bermudan, we can conclude that we would not cancel today.